Ogbonna Arinze

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SUMMARY

Lead Fullstack Engineer with 8 years of professional experience architecting production-grade React and Next.js applications with real-time API systems, robust CI/CD infrastructure, and scalable backend services. Proven track record of migrating prototypes to production at scale across healthcare, fintech, and civic tech, with deep expertise in TypeScript, NestJS, PostgreSQL, and real-time data streaming (SSE, WebSockets, EventSource). Skilled at building design systems, establishing testing frameworks (Playwright, Jest), and leading technical direction while mentoring engineering teams. Experienced in AI/LLM integration, having shipped production RAG systems and multi-agent pipelines.

CORE COMPETENCIES

- **Fullstack Architecture:** React, Next.js 14, TypeScript, NestJS, NodeJS, monorepos (Nx, pnpm), SSR/SSG, design systems, and scalable component libraries.
- **Real-Time APIs & Data:** REST APIs, GraphQL, SSE streaming, WebSockets, EventSource API, PostgreSQL, MongoDB, Prisma, and low-latency data pipelines.
- **CI/CD & Infrastructure:** Docker, Docker Compose, GitHub Actions, AWS (CloudFront, S3), Vercel, Heroku, multi-stage Dockerfiles, health checks, and automated deployment pipelines.
- **Testing & Quality:** Playwright (E2E), Jest, Cypress, Allure Reports, Storybook, ESLint, Prettier, and CI-driven QA with automated reporting.
- **AI & LLM Systems:** RAG pipelines, OpenAI API, LangChain, Vercel AI SDK, Mastra, vector search (pgvector, Chroma), embeddings, and multi-agent workflows.
- **Performance & Observability:** Code splitting, lazy loading, CDN strategies, Lighthouse, Core Web Vitals, bundle analysis, Sentry, and Langfuse.
- **Leadership:** Technical direction, architecture reviews, mentorship (4+ engineers mentored), Agile/Scrum, and product-minded development.

WORK EXPERIENCE

Genentech (Roche) San Francisco, CA (Remote)
Frontend Engineer *May 2023 - Present*

- Spearheaded the redesign of the main clinical trial data product, migrating from a **Vue.js** web application to a **React Chrome extension**, architecting the production-grade frontend from prototype to scalable deployment.
- Co-developed a full-stack **Next.js chat widget** powered by a **RAG system** using **OpenAI GPT models**, **LangChain**, **Vercel AI SDK**, and **Chroma** vector database, enabling real-time conversational data querying across connected data sources.
- Architected a **reusable design system** in **React** and **Radix UI** with **Tailwind CSS**, improving UI consistency, accessibility (WCAG), and developer velocity across multiple product teams.
- Implemented **JWT-based authentication** and **role-based access control** in **Next.js** with **TypeScript** and **React Query**, using HTTP-only cookies for secure token handling and permission-based UI rendering.
- Optimized application performance by implementing **lazy loading, code splitting, dynamic imports**, and **CDN asset delivery**, reducing load times significantly as measured via **Lighthouse** and Chrome DevTools.
- Introduced **Redux** for state management and **React Query** for request caching, refactoring legacy class components to functional components and streamlining data flow across the application.
- Increased release reliability by integrating **Playwright end-to-end tests** into the **CI/CD pipeline**, automating reporting with **Allure Reports** on GitHub Pages and configuring Slack notifications for build quality visibility.
- Mentored a team of two junior developers (interns), fostering a culture of technical excellence, clean code practices, and thorough code reviews.

Descasio Lagos, Nigeria (Hybrid)
Senior Frontend Developer *May 2021 - Mar 2023*

- Led the design and implementation of an intuitive **React** user interface, achieving a **25% boost in user engagement** and collaborating with architects to create a scalable frontend architecture that accelerated development by **30%**.
- Optimized frontend performance by profiling render bottlenecks with **React Profiler** and **Lighthouse**, reducing bundle size by **30-40%** through rollup-plugin-visualizer and webpack-bundle-analyzer, achieving a **35% faster load time**.
- Drove a **40% improvement** in application performance by optimizing caching mechanisms across the React application.
- Enhanced development workflow by introducing **Jest** and **Enzyme** testing frameworks, leading to a **20% decrease** in bug reports.
- Established engineering standards using **ESLint, Prettier, Husky, Lint-staged**, and **GitHub Actions**, reducing errors and accelerating CI/CD processes across the team.

- Utilized **Storybook** for documenting React components, streamlining change management and supporting agile development practices.
- Mentored junior engineers on frontend best practices, fostering high-quality, maintainable code across projects.

ActivEdge Technologies Lagos, Nigeria
Senior React Developer *Oct 2019 - Sep 2021*

- Built a high-volume **bank Settlement Solution** in a team of 3, handling thousands to millions of daily transactions and boosting settlement rates by **80%**.
- Developed critical payment modules (MasterCard, VISA, Express-cash, E-commerce) using **React** with **Redux** for state management, **RxJS** for reactive data streaming, and **Axios** for asynchronous API requests.
- Built a real-time account reconciliation solution, boosting daily reconciliation by **60%** through innovative transaction processing and data management strategies.
- Adopted the native **EventSource API** for **real-time data streaming**, replacing socket.io for faster, lower-latency updates and a more responsive interface.
- Built dynamic data visualization dashboards using **React** and **react-chartjs** with **Redux Toolkit** and **Ant Design** for user role and permission management.

Founders Lair Beijing, China (Remote)
Full Stack Engineer (Contract) *Oct 2020 - Feb 2021*

- Developed **NodeJS RESTful APIs** and translated business requirements into efficient **MongoDB** database models and schemas, collaborating closely with the business team to ensure technical solutions met product needs.
- Transformed Figma designs into reusable **React** components and engineered a **CI/CD pipeline** for automated deployment to **Heroku** (testing) and **AWS** (production).
- Mentored two interns through frontend and backend projects, providing technical guidance and structured task management.

SureTrade Business Solutions Nigeria
Frontend Developer *Mar 2018 - Oct 2019*

- Designed **MongoDB** database schemas and implemented server-side logic in **NodeJS**, building and testing **REST API** endpoints for multiple client-facing applications.
- Developed **SureEdu**, a school management portal in a team of 5, handling staff payroll and attendance tracking, student records (exams, reports, attendance), and guardian communications.
- Responsible for full-stack integration, connecting the **JavaScript** frontend to **NodeJS** backend services, developing user use cases, and converting business requirements into database schemas and models.
- Devised comprehensive documentation for each application, detailing functions, capabilities, and features. Tested user-defined functions, triggers, and API endpoints to ensure reliability.

PROJECTS

OurNigeria ournigeria.ng
Senior Engineer *Feb 2026 - Present*

AI-powered Nigerian budget analysis and corruption tracking platform. 700+ budget documents, 708K+ vector embeddings, 37 states.

- Architected the entire platform as an **Nx monorepo** managing **6 applications**: **NestJS** API backend, **Next.js** citizen-facing web app (React 19, Tailwind CSS v4, 22 interactive chart types with Recharts), Next.js admin dashboard, NestJS document ingestion pipeline, Next.js marketing landing page, and a Remotion video generation service.
- Implemented **SSE streaming** for real-time chat responses with 15s keepalive heartbeats to prevent reverse proxy timeouts, token-budgeted context management, and conversation summarization, demonstrating deep proficiency in **low-latency real-time data delivery**.
- Built a **multi-agent AI system** using **Mastra** with 5 specialized agents and a three-stage intent classification pipeline, powered by a **hybrid RAG pipeline** combining **pgvector** + BM25 full-text search with Reciprocal Rank Fusion and **Cohere** reranking.
- Designed the **PostgreSQL 16** database schema with **Prisma v7**, including pgvector extension for vector storage, and built custom migration tooling that filters out framework-managed tables from drift detection.
- Configured **CI/CD** with **GitHub Actions** for automated builds, **Playwright** E2E tests across 4 projects, and **Docker Compose** production deployment with multi-stage Dockerfiles, health checks, memory limits (API 1GB, Ingest 4GB, Postgres 2GB), and **Traefik** reverse proxy routing.

EDUCATION

University of Nigeria Nsukka, Enugu
BSc Computer Science - 4.1 CGPA *Graduation Date: Sep 2019*